

## EQUATIONS AND INEQUALITIES

From Grade 11 onwards the majority of equations that you have to solve will be quadratic.

### 1. Quadratic Equations

These are recognisable by the 'square' with the variable – it will be the highest power  
 $\therefore$  2 solutions!

Steps to solving:

- All terms one side and zero on the other
- Factorise fully
- Make each factor equal to zero and solve for both solutions

### 2. Equations with fractions

- Make sure all denominators are fully factorised
- State the restrictions on the variable (can't have the denominator = 0)
- Find LCD
- Multiply each term throughout equation by LCD (you should have no fractions left after this step)
- Remove brackets, collect like terms and solve. (If quadratic, follow steps above)

### 3. Using substitution to solve equations (often referred to as the k-method)

If a  $\frac{2}{3}$  term expression is repeated in more than one place in an equation, this method is suitable to help simplify the equation in order to solve it.

Steps to solving:

- State the restrictions on the variable
- Let the repeated expression equal 'k'
- Solve the equation using 'k' in the expression's place.
- Once solved, replace k by the original expression and solve again.

These equations usually have more than 2 solutions.

4. Simultaneous Equations (Given 2 equations with 2 variables to solve for)

4.1 Substitution method

- Make ONE of the variables the subject of the equation ONE of the equations
- Use this information to substitute back into the second equation. There should now be an equation with only one unknown variable in it
- Solve for this variable
- Use the information just found to substitute back into the first equation and solve for the second variable.

4.2 Elimination method (works well if one of the variables has the same co-efficient in each of the 2 equations)

- Place the 2 equations below each other ensuring all the same terms are underneath each other
- Subtract the one equation from the other (if the co-efficients have the same sign) in order to eliminate that variable

OR

Add the 2 equations together (if the co-efficients have opposite signs) in order to eliminate one of the variables

- Solve the 'new' equation which has only one unknown
- Use the information just found to substitute back into one of the original equations to solve for the missing variable.

5. Solving equations by squaring both sides

When an equation has a square root in it, the only way of solving is to square it.

Steps to solving:

- Get the term with the square root on its own on one side
- Square both sides (If there is more than one term on the other side,  
BEWARE!... $(x + 3)^2 \neq x^2 + 9$  !!!)
- Solve as per usual

However: Each solution needs to be checked. Squaring can often lead to an invalid solution.

Substitute each solution into the original LHS and RHS of the equation to check that LHS = RHS.

If so, the solution is valid

If not, that particular solution is not valid.

6. Solving by completing the square

- The aim is to form a perfect square trinomial on the LHS
- To do this, put all terms with variables on LHS and constants on RHS
- Complete the square on LHS
- Factorise the perfect square trinomial
- Square root both sides and solve. Remember both solutions (+/-) when square rooting

7. Solving by using the Quadratic formula

- Get all terms on one side (equal to zero)
- Collect like terms to get into the form:  $ax^2 + bx + c$
- Substitute into  $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$  to solve

8. Quadratic Inequalities (2 methods)

- Make RHS = 0
- Factorise LHS fully
- Find critical values (In general, there will be one per factor)
- Put the cv's on a numberline
- Place open ( > or < ) or closed (  $\leq$  or  $\geq$  ) dots on each number (cv's)
- 8.1 (Testing all areas for valid solutions)
- Test all 'areas'(sections) for positive or negative answers (by substituting in any value belonging in that area)
- (Number of areas should be number of critical values + 1)
- Choose appropriate answer according to given inequality.
- 8.2
- Sketch a parabola using the critical values as the x-intercepts on the numberline
- Highlight the positive or negative part of the parabola according to the given inequality (< or >)

Show solution as per question (Numberline, Interval notation or as an Inequality)